Using Geostationary Satellite Ocean Color Data to Map Diurnal Hourly Velocity Field Changes in Oceanic Mesoscale Eddy

Xiaosong Ding, Xianqiang He, Yan Bai, Jiajia Li, and Fang Gong

***Abstract*— The mesoscale eddy structure in the upper ocean is difficult to observe with altimeter satellite data due to their coarse spatiotemporal resolution. However, the world's first Geostationary Ocean Color Imager (GOCI) makes it possible to refine the eddy structure inversion with observations of high spatiotemporal resolution (500 m and 1 h, respectively). In this study, we proposed a new flow field retrieval algorithm recurrent all-pairs field transforms in mesoscale eddy flow retrieval (RAFT-MesoEddy) based on deep current optical flow architecture in the mesoscale eddy surface flow field for GOCI data. The validation results showed that the flow field matched the particle image velocimetry (PIV) data in both quality and spatial distribution, with mean end-to-end point error (MEPE) and mean average angle error (MAAE) of less than 0.07 pixel and 2.57°, respectively. The GOCI-derived flow field showed good consistency with drifting buoy trajectory data, with MEPE and MAAE values less than 0.34 m/s and 18.78°, respectively. The GOCI-retrieved flow field showed agreement with satellite altimeter geostrophic flow data, with the MEPE and MAAE being 0.28 m/s and 10.12°, respectively. The GOCI-retrieved flow field showed an agreement with Ocean Surface Current Analyses Real-time (OSCAR) surface flow field, with the MEPE and MAAE are 0.28 m/s and 9.62°, respectively. In addition, a specific mesoscale eddy to discuss the hourly dynamics of flow field and eddy kinetic energy (EKE) is taken. The significant diurnal dynamics revealed in the hourly GOCI observations suggest caution in mapping mesoscale eddy flow using conventional altimeter satellite data, as the temporal resolution is insufficient to capture diurnal variations.**

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Xiaosong Ding is with Donghai Laboratory, Zhoushan 316021, China, and also with the Ocean College, Zhejiang University, Zhoushan 316021, China (e-mail: d_xiaosong@163.com).

Xianqiang He and Yan Bai are with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou 310012, China, and also with the Ocean College, Zhejiang University, Zhoushan 316021, China (e-mail: hexianqiang@sio.org.cn; baiyan@sio.org.cn).

Jiajia Li is with the Institute of International Rivers and Eco-Security, Yunnan University, Kunming 650500, China (e-mail: l_jiajia1996@163.com).

Fang Gong is with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou 310012, China (e-mail: gongfang@sio.org.cn).

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***Index Terms*— Diurnal change, flow field retrieval, Geostationary Ocean Color Imager (GOCI), mesoscale processes, optical flow.**

I. Introduction

OCEANIC mesoscale eddies cover 25%–30% of the ocean surface areas and transport over 90% of the material and energy in the global oceans, and the eddy-induced regional mass transport is of a magnitude comparable to that of the large-scale circulation driven by the wind and the thermohaline circulation [1], [2], [3]. However, their impact on the global ocean energy budget carries substantial uncertainty [4], [5], [6]. Satellite altimeter data, drifting buoy trajectory data, and model data were the primary data sources for investigating oceanic mesoscale processes. Altimeter satellite data and its fused datasets offer daily global operational observations of sea surface dynamic processes; for example, Archiving, Validation and Interpretation of Satellite Oceanographic data (AVISO) has provided important daily observation with a spatial resolution of 0.25° (25 × 25 km) since 1992 (https://www.aviso.altimetry.fr). However, its coarse spatial resolution inevitably results in the loss of essential features in the mesoscale processes [7], [8], [9]. Drifting buoy trajectories are research data that use Lagrangian methods to track the spatial movement of water masses [10], [11]. Due to limitations in observation coverage, these data cannot effectively capture the detailed characteristics of large-scale spatial flow field variability [9], [12]. Model simulations, such as the Hybrid Coordinate Ocean Model, the Massachusetts Institute of Technology General Circulation Model (MITgcm), and the Nucleus for European Modelling of the Ocean were still affected by spatial resolution constraints, inadequate parameterization schemes, uncertainty in initial and boundary conditions, and a lack of understanding of complex dynamic processes, such as boundary layer turbulence, eddy mixing, and bottom friction, which may result in an inability to accurately describe complex mesoscale processes [13], [14], [15]. Currently, the study of oceanic mesoscale mass and energy transport is significantly hampered by the limitations of low spatial resolution and single daily observations [8], [16], [17].

In recent years, many physical oceanographers have been interested in geostationary ocean color products, such as chlorophyll-a (Chl-a) and suspended particulate matter (SPM) [8], [16], [18], [19]. These ocean color products can be used

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as tracer particles to indicate the high-frequency flow field variations of mesoscale processes in the near-surface layer [8], [9], [17]. Geostationary Ocean Color Imager (GOCI) was launched in June 2010 and covers the Korea/China/Japan region with up to 8 images/day (00:15–07:15, UTC time) with a spatial resolution of 500 m [18], [20], [21]. The maximum cross correlation (MCC) algorithm is the most commonly used method for the extraction of flow fields from GOCI data [9], [17], [19]. The MCC algorithm principle is to divide the image into small windows, compare their cross correlation in two consecutive time images, and ultimately determine the spatial displacement of water masses [9], [17], [22]. Hu et al. [9], [23] extracted M2 tidal flow fields and diurnal residual flow variations from GOCI-retrieved SPM products in the Yangtze River estuary and Hangzhou Bay. Based on the GOCI Chl-a product, Choi et al. [22] compared the effect of different matching window sizes on the accuracy of extracting flow fields for submesoscale turbulent processes in the Japan Sea/East Sea. Using GOCI Chl-a and 555 nm remote sensing reflectance (Rrs555) products, Liu et al. [17] extracted long-term flow field data in the East China Sea. Considering the curvature effect, Lee and Mei [24] proposed diffeomorphic particle image velocimetry (DiffeomorphicPIV) and applied it to improve flow field retrieval in rotationally distorted environments within the standard particle image velocimetry (PIV) algorithm. However, the MCC algorithm exhibited considerable uncertainty in retrieving velocity fields, especially in rotationally distorted flow fields. This uncertainty arises from subjective settings, including the search window size, the interrogation window size, and the determination correlation (R^2) threshold [17], [22], [23].

Different from the traditional MCC algorithm, the optical flow method proposed a dense optical flow architecture that uses pixel-based spatial units and assumes constant gray values between two consecutive images [25], [26], [27]. Dérian and Almar [28] used optical flow to extract a real-time surface flow field based on the detected motion of foam patterns in the oceanic near-surface layer. With the development of convolutional neural networks (CNNs) in the field of optical flow in computer vision, there are many high-precision deep learning architectures, such as optical flow estimation with deep networks (FlowNet2) [29], pyramid-warping-cost-volume networks (PWC-Net) [30], lightweight CNNs for optical flow networks (LiteFlowNet) [31], and recurrent all-pairs field transforms for optical flow networks (RAFT) [32]. The combination of laboratory laser PIV data has been used to improve particle optical flow fields in the above architecture, resulting in LiteFlowNet-PIV [33], UnLiteFlowNet-PIV [34], and RAFT-PIV [35]. Compared with LiteFlowNet-PIV and UnLiteFlowNet-PIV, RAFT-PIV is robust to different particle sizes, works well for signal-to-noise ratio >2, and performs best in high seeding conditions [35]. The RAFT-PIV architecture has notable advantages in extracting flow field, especially in the rotation-distorted velocity field [35], [36], [37]. However, there are few studies that apply this technique to the refined extraction of ocean surface flow field from geostationary satellite observational products.

Motivated by the opportunities and challenges in sea surface displacement velocity retrieval of highly rotated and distorted mesoscale and submesoscale dynamics, this study leverages high spatiotemporal resolution data from geostationary ocean color satellites and introduces a deep CNN architecture, RAFT in mesoscale eddy flow retrieval (RAFT-MesoEddy). This architecture aims to develop advanced remote sensing techniques for fine-scale displacement velocity retrieval with high temporal (1 h) and spatial (500 m) resolutions. The contents are organized as follows: Section II introduces the data sources and the model accuracy assessment methods. Section III provides the RAFT-MesoEddy architecture proposed in this study and its technical details. Section IV discusses the accuracies of different algorithms in inversing displacement velocity based on the public PIV dataset, the advantages of the RAFT-MesoEddy architecture in fine-scale inversion of complex flow fields, and the validations of the performance of the RAFT-MesoEddy architecture using Drifter trajectory, AVISO, and Ocean Surface Current Analyses Real-time (OSCAR) datasets. Finally, the applications of the RAFT-MesoEddy architecture to monitor hourly variations in eddy flow field and kinetic energy are demonstrated.

II. Data

A. PIV Datasets for Neural Network Training

The PIV dataset followed the general flow motion model to generate both a particle image and flow data [33]. In this study, nine types of PIV data (8.93 GB) were used for training (70%) and validation (30%) of the RAFT-MesoEddy architecture from https://github.com/shengzesnail/PIV_dataset (Table I) [33]. The two consecutive particle images build on the idea that a particle can be described by a 2-D Gaussian function

$$I_{(x_1,y_1)} = I_0 \exp\left[\frac{-(x_1-x_0)^2-(x_1-y_0)^2}{(1/8)d_p^2}\right] \quad (1)$$

where $I_{(x_1,y_1)}$ is the displacement estimations of the second frame from the first frame location I_0; (x_0, y_0) and (x_1, y_1) are the particle positions of the two consecutive images.

B. GOCI Chl-a Product

GOCI provided near-real-time hourly ocean color data from 00:15 to 07:15 (UTC time) with a spatial resolution of 500 m in regions covering Korea, Japan, and eastern China sea [21]. GOCI remote sensing reflectance in three bands (443, 490, and 555 nm) was used for Chl-a product retrieval using the OC3 algorithm with coefficients adjusted for the Korean region [8], [21], [38]. As the successor to GOCI, launched in 2020, GOCI-II provides higher spatial resolution data (250 m) from 23:30 to 08:30 (UTC time). In this study, we investigated the refined mesoscale process flow field inversion technique from GOCI/GOCI-II Chl-a products (Table II).

C. Validation Data

The trajectory data of drifters from the National Oceanic and Atmospheric Administration (NOAA) Global Drifter Program (https://www.aoml.noaa.gov/phod/gdp/hourly_data.php)

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TABLE I
PIV TRAINING DATA AND VALIDATION DATA IN THIS STUDY

Types	Training dataset		Validation dataset		Shape (Col × Row)	Description
	Number (F1 × F2 × Flow)	Size (GB)	Number (F1 × F2 × Flow)	Size (GB)		
Backstep	2520	1.344	1080	0.576	256×256	Backward stepping flow
Cylinder	4305	0.798	1845	0.342	256×256	Flow over a circular cylinder
DNS turbulence	4200	0.84	1800	0.36	256×256	A homogeneous and isotropic turbulence flow
Johns Hopkins Turbulence Databases (JHTDB) hd	4689	0.798	2010	0.342	256×256	Turbulent channel flow database
JHTDB channel	1260	0.252	540	0.108	256×256	Channel flow provide by JHTDB
JHTDB isotropic1024	4200	0.84	1800	0.36	256×256	Forced isotropic turbulence provide by JHTDB
JHTDB mhd1024	1677	0.329	720	0.141	256×256	Forced MHD turbulence provide by JHTDB
Surface Quasi Geostrophic (SQG) Turbulence	3150	0.63	1350	0.27	256×256	Sea surface flow by SQG model
Uniform	4200	0.42	1800	0.18	256×256	Uniform flow field
SUM	30201	6.251	12945	2.679	256×256	

TABLE II
GOCI AND GOCI-II IMAGE DATA WERE USED IN THIS STUDY

Types	Temporal resolution	Spatial resolution	Diurnal observation frequency	Date
GOCI	1 h	500 m	8 (00:30 — 07:30, UTC)	2011/04/05 2019/05/03 2019/05/04 2019/05/25
GOCI-II	1 h	250 m	10 (23:30 — 08:30, UTC)	2022/01/03 2022/04/04 2022/09/11 2023/06/05

were obtained through satellite tracking of surface drifting buoys. The Drifter Data Assembly Center at NOAA's Atlantic Oceanographic and Meteorological Laboratory applies quality control processes and data processing to enhance these observations, producing hourly estimates of sea surface temperature and flow field data. The dataset includes latitude and longitude positions, sea surface temperatures (including total, diurnal, and nondiurnal components), velocities (eastward and northward, unit: m/s), and associated uncertainty assessments. In this study, two drifters selected on April 5, 2011, in the southern waters of the Japan Sea/East Sea (131.11° E, 35.81° N) and in the coastal waters of southern Shikoku, Japan (133.76° E, 32.84° N) were used to validate the accuracy of the GOCI retrieval flow field. In addition, trajectories from four long-term buoys in the Japan Sea (from April 1 to April 30, 2016) were also used to verify the accuracy of the GOCI retrieval flow field. The time difference should not exceed ±30 min, and the mean flow value in a box (3 × 3 pixels) central to the drifter buoy position was adopted.

OSCAR surface flow field provides daily 25-km resolution global ocean geostrophic flow fields since 1993 [39] (https://catalog.data.gov/dataset/ocean-surface-current-analyses-real-time-oscar-surface-currents-final-0-25-degree-version–29f4f). OSCAR ocean mixed-layer velocities were computed using a simplified physical model for geostrophic, Ekman, and thermal wind dynamics from satellite-derived sea surface height gradients, ocean vector winds, and sea surface temperature gradients. For comparison with the OSCAR data, the GOCI-retrieved sea surface flow field has been downsampled from 500 m to 25 km spatial resolution, which can be used to compare the spatial distribution consistency with the OSCAR surface flow field.

The AVISO data provide daily 25-km resolution global ocean geostrophic flow fields since 1992. For comparison with the AVISO data, the GOCI-retrieved sea surface flow field has been downsampled from 500 m to 25 km spatial resolution, which can be used to compare the spatial distribution consistency with the AVISO geostrophic flow field.

For GOCI-II Chl-a products, the Korea National Ocean Satellite Center (NOSC) has released surface flow field products based on the MCC algorithm [40], [41]. The NOSC flow field products (Table II) were also used to validate the robustness of the sea surface flow algorithm in this study.

D. Accuracy Assessment

The end-to-end point error (EPE) is calculated by comparing an estimated optical flow vector (u', v') with a reference flow vector (u, v), and the EPE is defined as the Euclidean distance between these two flow fields [29], [35]. The mean EPE (MEPE) is defined as the mean of the EPE values in all the pixels. The formulas for the EPE and the MEPE are as

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follows:

$$\mathrm{EPE}_{(i,j)} = \sqrt{\left(u_{(i,j)} - u'_{(i,j)}\right)^2 - \left(v_{(i,j)} - v'_{(i,j)}\right)^2} \quad (2)$$

$$\mathrm{MEPE} = \frac{1}{N} \times \sum_{i=0}^{i=m} \sum_{j=0}^{j=n} \mathrm{EPE}_{(i,j)} (N = m \times n). \quad (3)$$

The average angular error (AAE) [9], [42], quantified in degrees, is calculated for estimated flow and reference flow. It calculates the mean AAE (MAAE) of the angular distance between two directions. The vectors are transformed into Cartesian unit vectors, and their scalar product gives the cosine of the angle between them. The inverse cosine (arccos) of this value represents the angle between the input flow field. The formulas for the AAE and the MAAE are as follows:

$$\mathrm{AAE}_{(i,j)} = \arccos\left(\frac{u_{(i,j)}u'_{(i,j)} + v_{(i,j)}v'_{(i,j)}}{\sqrt{u'^2_{(i,j)}v'^2_{(i,j)} + 1}\sqrt{u^2_{(i,j)}v^2_{(i,j)} + 1}}\right) \quad (4)$$

$$\mathrm{MAAE} = \frac{1}{N} \times \sum_{i=0}^{i=m} \sum_{j=0}^{j=n} \mathrm{AAE}_{(i,j)} (N = m \times n). \quad (5)$$

III. Methodology

The RAFT-MesoEddy architecture for the extraction of oceanic mesoscale process surface dynamic flow field combined the standard PIV algorithm and the RAFT optical flow architecture. The details are as follows (Fig. 1).

A. Standard PIV Algorithm

The standard PIV algorithm is an image processing and flow field retrieval technique that uses the MCC algorithm computed to track tracer particles spatial displacement from two consecutive images [8], [43]. In Step 1 of Fig. 1, the search window size and the overlap window size are 20 × 20 pixels and 8 × 8 pixels, respectively. I flow field is predicted in Step 1, and the standard PIV algorithm outputs a spatially sparse displacement field and requires postprocessing.

B. RAFT Architecture

The RAFT optical flow architecture consisted of several key components to achieve efficient and precise optical flow estimation (Fig. 1, Step 2) [30], [32], [35]. First, the feature extraction phase identifies important features from images, which exhibit good stability and robustness across different frames of data. Second, the feature matching phase establishes correspondences between feature points, providing input data for subsequent optical flow calculations. Third, in the optical flow estimation step, the architecture employed CNNs to simulate the motion patterns of feature points, enabling accurate estimation in complex scenes. The multiscale strategy allows optical flow estimation at different scales to adapt to varying motion and changes. Finally, the filtering and iteration steps improve the estimation accuracy and robustness through iterative optimization and smoothing processes, which are particularly effective for boundary and rotationally distorted regions.

In this study, the RAFT architecture was trained using a publicly available measured PIV dataset (Table I) [33]. The epoch is 120 000, the optimizer is Adam, the learning rate is 0.0001, the pretrained checkpoint is 5000, and the loss selection is endpoint error [see (2)]. Compared to the flow field derived by the standard PIV algorithm in Fig. 1 (Step 1), the flow field predicted in Step 2 using the RAFT architecture was significantly clearer and smoother, effectively retrieving the flow velocity from two consecutive images.

C. Oceanic Mesoscale Process Flow Field Retrieval

The flow field predicted by the RAFT-MesoEddy architecture must be transformed according to the tempospatial resolution of the GOCI as follows:

$$\begin{cases} U_{\mathrm{GOCI}} = U_{\mathrm{RAFT}} \times \dfrac{\mathrm{SR}}{\mathrm{TR}} \\ V_{\mathrm{GOCI}} = V_{\mathrm{RAFT}} \times \dfrac{\mathrm{SR}}{\mathrm{TR}} \end{cases} \quad (6)$$

where U_{GOCI}, V_{GOCI}, and U_{RAFT} and V_{RAFT} are the horizontal and vertical velocities (unit: m/s) of GOCI and RAFT, respectively. SR in GOCI (GOCI: 500 m/pixel; GOCI-II: 250 m/pixel) and TR (unit: 3600 s) are the spatial and temporal resolutions, respectively.

D. Diurnal Mesoscale Process Change

In Step 4 of Fig. 1, a refined extraction of sea surface flow fields with higher tempospatial resolution (500 × 500 m, 1 h, respectively) was performed based on the diurnal hourly GOCI Chl-a products and the RAFT-MesoEddy architecture. To ensure the robustness of the extracted flow fields, four typical mesoscale eddies are selected to examine the stability and consistency of the diurnal hourly pixel-level flow field.

IV. Results and Discussion

A. Assessment of the PIV Flow Field

Table III shows the accuracy based on 30% validation data for DiffeomorphicPIV, UnliteflowNet-PIV, and our proposed model. Overall, the RAFT-MesoEddy architecture performed better than other deep learning architectures, with MEPE less than 0.23 pixel. The UnLiteFlowNet-PIV architecture showed relatively good consistency with the reference flow field, while the DiffeomorphicPIV showed the lowest consistency with the reference flow field. The DiffeomorphicPIV algorithm generated a deformation field from the dense optical flow field predicted using the Gunnar Farneback algorithm [24], [44] and showed a relatively lower MEPE compared to the UnLiteFlowNet-PIV architecture.

Fig. 2 shows the accuracy of particle displacement prediction by three deep learning architectures. The RAFT-MesoEddy architecture can better predict particle displacement in complex flow fields, with MEPE and MAAE values less than 0.07 pixel and 2.57°, respectively. For MEPE and MAAE, our model also showed the best performance, followed by the UnLiteFlowNet-PIV architecture and finally by the DiffeomorphicPIV algorithm.

In the boundary areas of particle displacement [black rectangle area in Fig. 2(b) and (c)], the DiffeomorphicPIV algorithm

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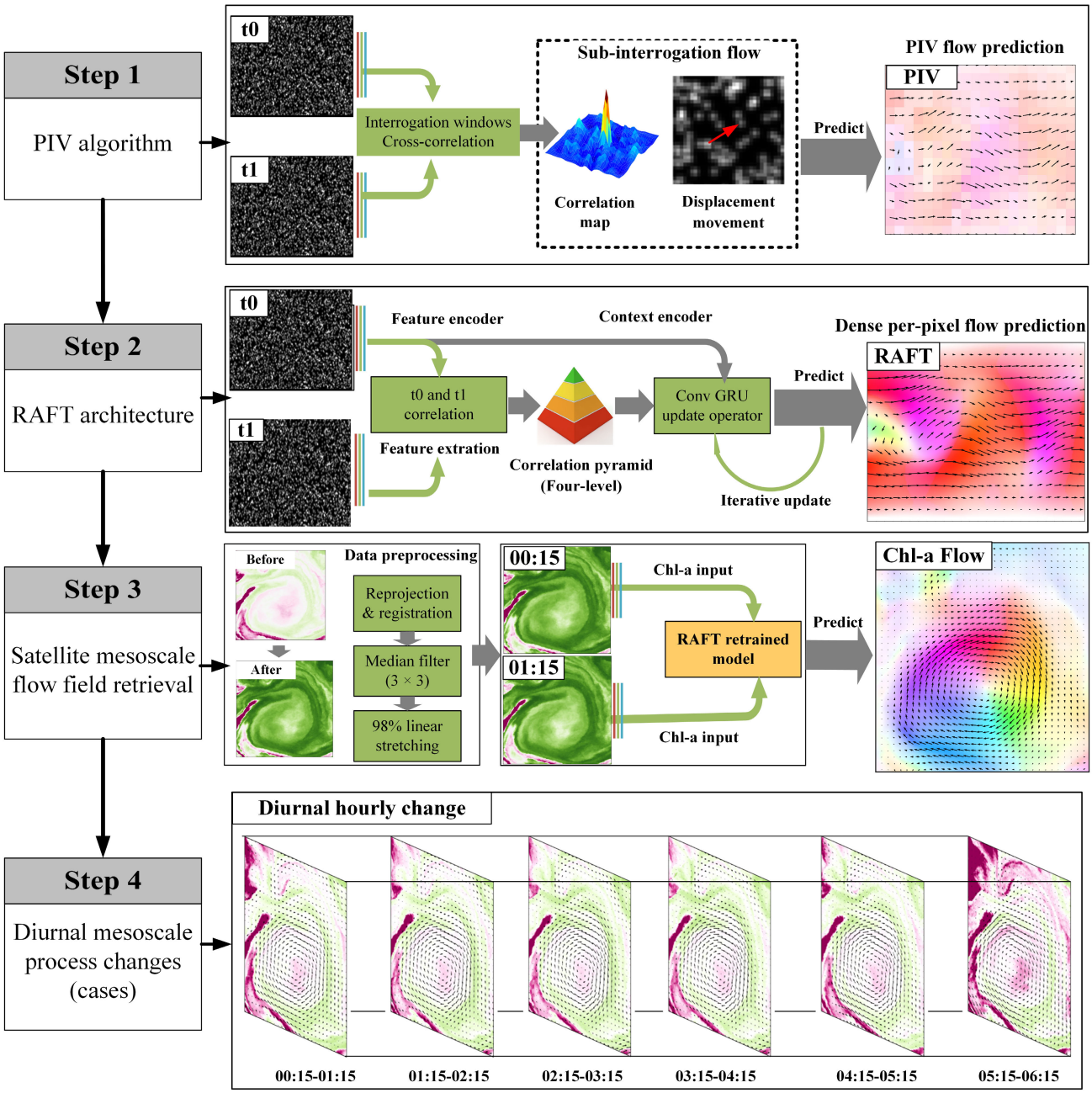


Fig. 1. Workflow of the RAFT-MesoEddy architecture. Step 1: The standard PIV algorithm architecture was introduced to extract the flow field of the PIV dataset. Step 2: The RAFT optical flow architecture was described to extract the flow field. Step 3: The oceanic surface flow field extraction framework was introduced by using the geostationary satellite GOCI diurnal hourly Chl-a product and RAFT-MesoEddy architecture. Step 4: The diurnal hour oceanic mesoscale process velocity field was extracted and discussed.

TABLE III
MEPE VALUE FOR THE TEST DATA

Methods	Backstep	Cylinder	JHTDB Channel	JHTDB Mhd1024	SQG
DiffeomorphicPIV	0.087	0.133	0.145	0.179	0.727
UnLiteFlowNet-PIV	0.108	0.060	0.095	0.145	0.240
Our results	0.086	0.050	0.071	0.139	0.226

and UnLiteFlowNet-PIV architecture were ineffective in eliminating the influence of boundary effects on flow field, while our model performs best in the boundary regions. In the missing and mixed pixel boundary areas caused by the cloud cover and the strong absorbing aerosol effect [16], [45], [46], the uncertainty of the flow field retrieval could be minimized by the RAFT-MesoEddy architecture. Therefore, our study attempts to apply the model to retrieve the sea surface flow field from the GOCI Chl-a product in the mesoscale eddies.

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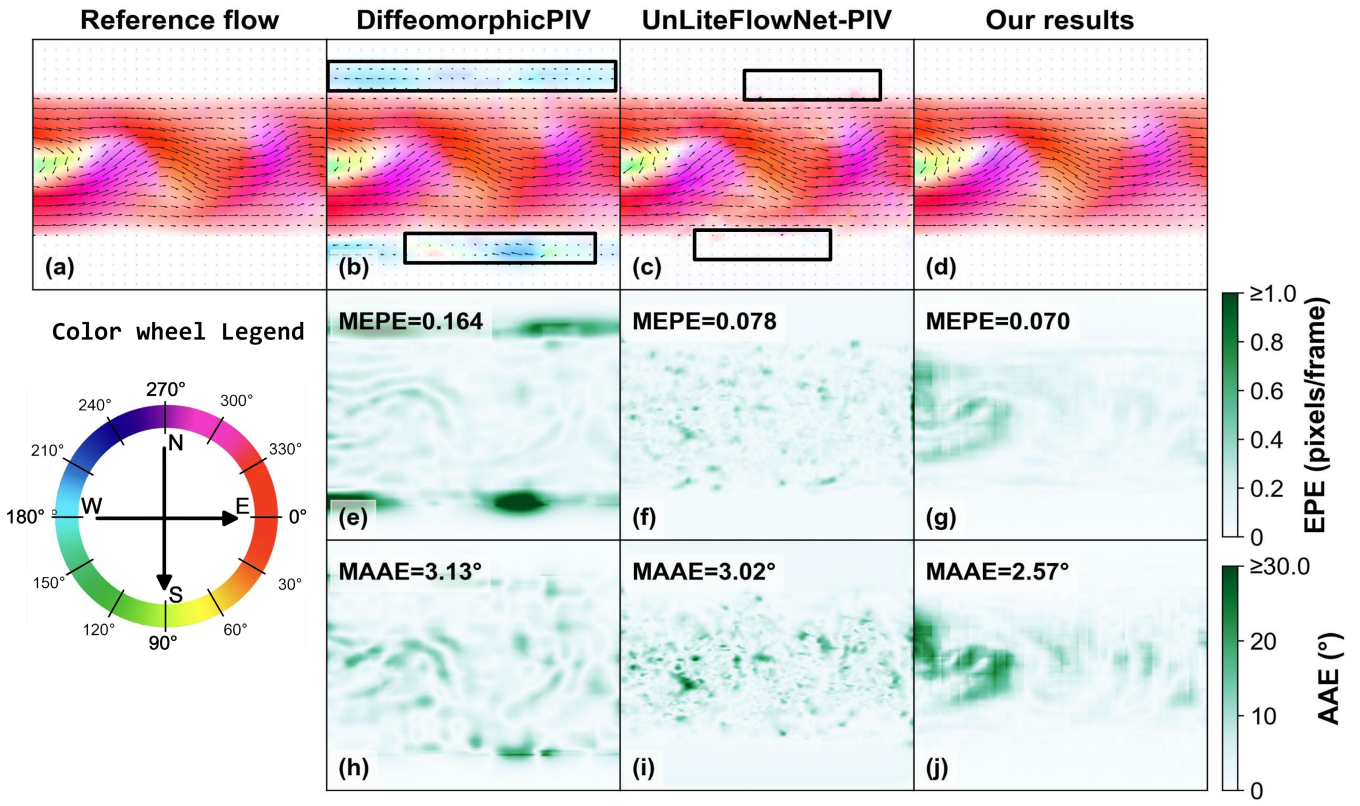


Fig. 2. Comparison of accuracy for different deep learning architectures. The black rectangle indicates the region with anomalous data. (a) Reference flow field. (b)–(d) Illustrate the flow field retrievals by DiffeomorphicPIV, UnLiteFlowNet-PIV, and RAFT-mesoscale architectures, respectively. (e)–(g) Depict the MEPE distribution for DiffeomorphicPIV, UnLiteFlowNet-PIV, and RAFT-mesoscale architectures, respectively. (h)–(j) MAAE distribution for DiffeomorphicPIV, UnLiteFlowNet-PIV, and RAFT-mesoscale architectures, respectively.

B. Surface Flow Retrieval of Oceanic Mesoscale Processes

The filamentous spatial structure of the oceanic surface phytoplankton is sensitive to mesoscale dynamic transport, which can be used as a tracer particle for flow field inversion [8], [22], [47]. With the GOCI Chl-a data, it is possible to retrieve the diurnal hourly flow fields [40], [48]. Compared with the standard PIV algorithm [Fig. 3(b)] [8] on April 5, 2011, the RAFT-MesoEddy architecture effectively captured the characteristics of the flow field variations in complex dynamic environments, including mesoscale eddies, fronts, and convection [Fig. 3(c)]. We have zoomed in on the surface flow field of the mesoscale eddies in the blue box [Fig. 3(c)]. The continuity of hourly changes in eddy boundaries and center positions was clear, highlighting the significant advantage of this algorithm in retrieving complex flow fields associated with mesoscale processes [Fig. 3(d)–(i)]. In the RAFT-MesoEddy architecture, the flow field matched well with the spatial structure of phytoplankton filaments, and both the flow fields and the GOCI image share the same spatial resolution (500 × 500 m). However, the traditional PIV algorithm did not perform well in the mesoscale eddy regions (Fig. 3), which could be attributed to the subjective nature of interrogating and selecting the search window [22], [23], [49]. Similarly, Choi et al. [8] extracted the surface flow field in the Japan Sea/East Sea using the standard PIV algorithm, but their spatial resolution did not achieve significant breakthroughs compared to altimeter satellite data in submesoscale turbulence. These coarse spatial resolution products still limit further understanding of the ocean surface mass and energy transport to mesoscale and submesoscale processes [50], [51].

The NOSC used the MCC algorithm to derive operational surface flow data from the GOCI-II satellite Chl-a products [40], [41], [52], [53]. These products are important datasets for understanding mesoscale processes in the oceanic near-surface layer [8], [54], [55]. To demonstrate the transferability and robustness of the RAFT-MeosoEddy architecture proposed in this study within GOCI-II imagery, we selected four days (January 3, 2022 representing Winter; April 4, 2022 representing Spring; June 5, 2023 representing Summer; and September 11, 2022 representing Autumn) of surface flow field products released from the official website of the Korea NOSC in Fig. 4. Overall, the NOSC surface flow field showed good agreement with the spatial structure of the filamentous phytoplankton, especially in the advective transport of water masses [Fig. 4(b)]. However, the highly rotated and distorted mesoscale and submesoscale flow fields showed numerous spatial patches of flow vectors [Fig. 4(h) and (k)]. The MCC algorithm struggles to accurately capture the detailed rotational and deformational motions in a flow field. Consequently, it often generates incorrect and disordered vectors that fail to represent the true motion of the sea surface flow field [54], [56].

The concentration of Chl-a exhibited periodic variations spanning from hourly to annual scales [17], [57]. Therefore, it is imperative to examine the spatial distribution patterns of varying Chl-a concentrations across different seasons to address uncertainties associated with the RAFT-MesoEddy

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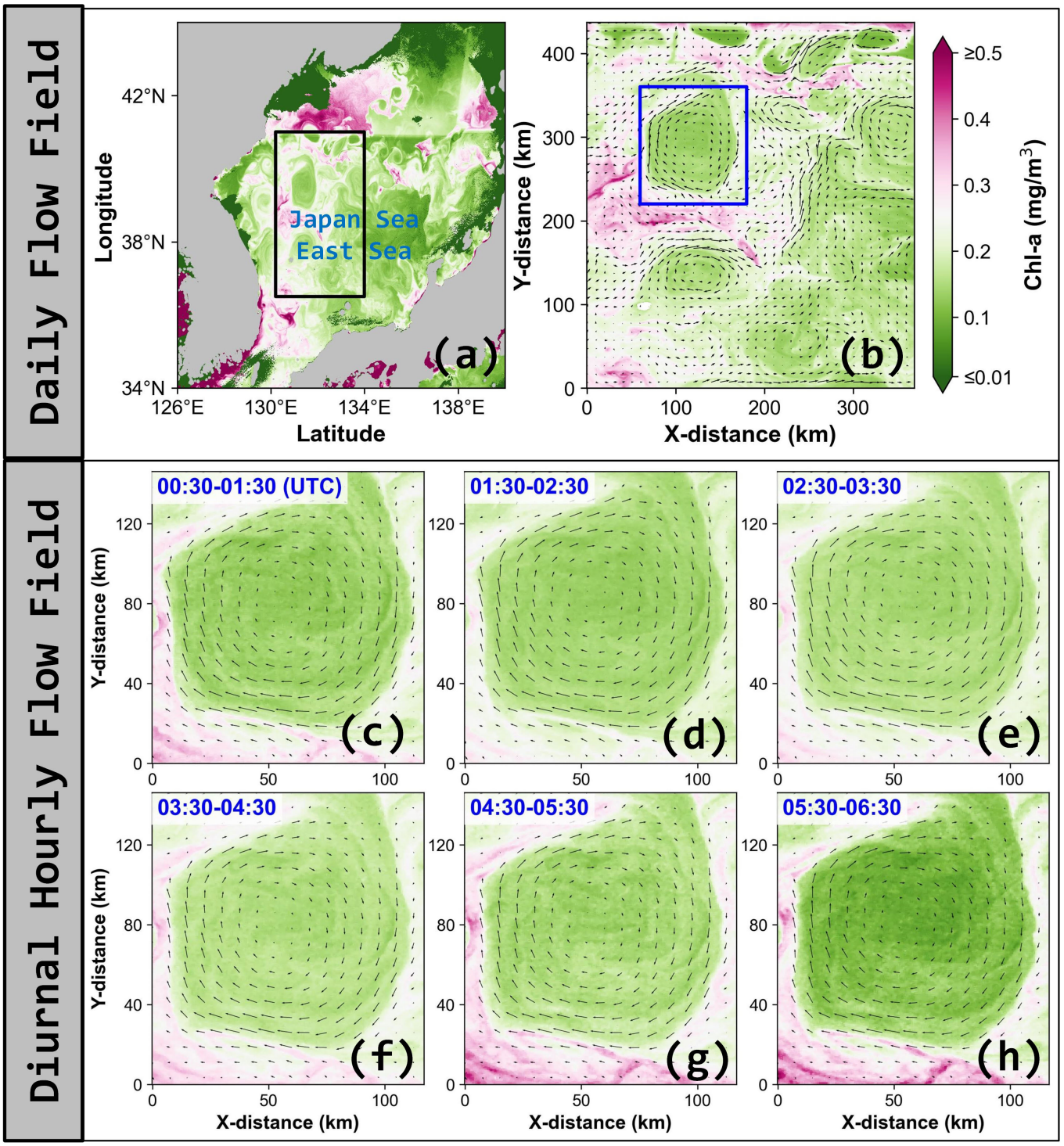


Fig. 3. Daily and diurnal hourly variations of mesoscale eddy flow fields retrieved using GOCI Chl-a products on April 4, 2011, comparing the standard PIV algorithm and our deep learning RAFT-MesoEddy architecture. (a) Study area (black rectangle) in the Japan Sea/East Sea, (b) daily flow field derived from the PIV algorithm, (c) daily flow field derived from the RAFT-MesoEddy architecture, and (d)–(i) diurnal hourly surface flow fields in the zoomed area (blue rectangle) from (c).

architecture. In winter, when the spatial gradients of Chl-a concentration were unclear, the NOSC MCC algorithm exhibited reduced accuracy in extracting flow field vectors [Fig. 4(b)]. In contrast, the RAFT-MesoEddy architecture effectively retrieved flow fields consistent with the background filamentous structure of phytoplankton [Fig. 4(c)]. In spring, with clearer spatial gradients of Chl-a concentration, both the NOSC flow field [Fig. 4(e)] and our results [Fig. 4(f)] accurately reflected the phytoplankton filamentous structure. However, NOSC data still showed errors in the mesoscale eddy frontal region. In summer, despite the continued clarity in spatial gradients of Chl-a concentration, the flow vectors from NOSC were even less precise and appeared spatially patchy [Fig. 4(h)]. Conversely, the RAFT-MesoEddy architecture maintained effective retrieval of flow fields consistent with the phytoplankton filamentous structure [Fig. 4(i)]. In autumn, the NOSC flow field demonstrated poor spatial continuity, inconsistent with the

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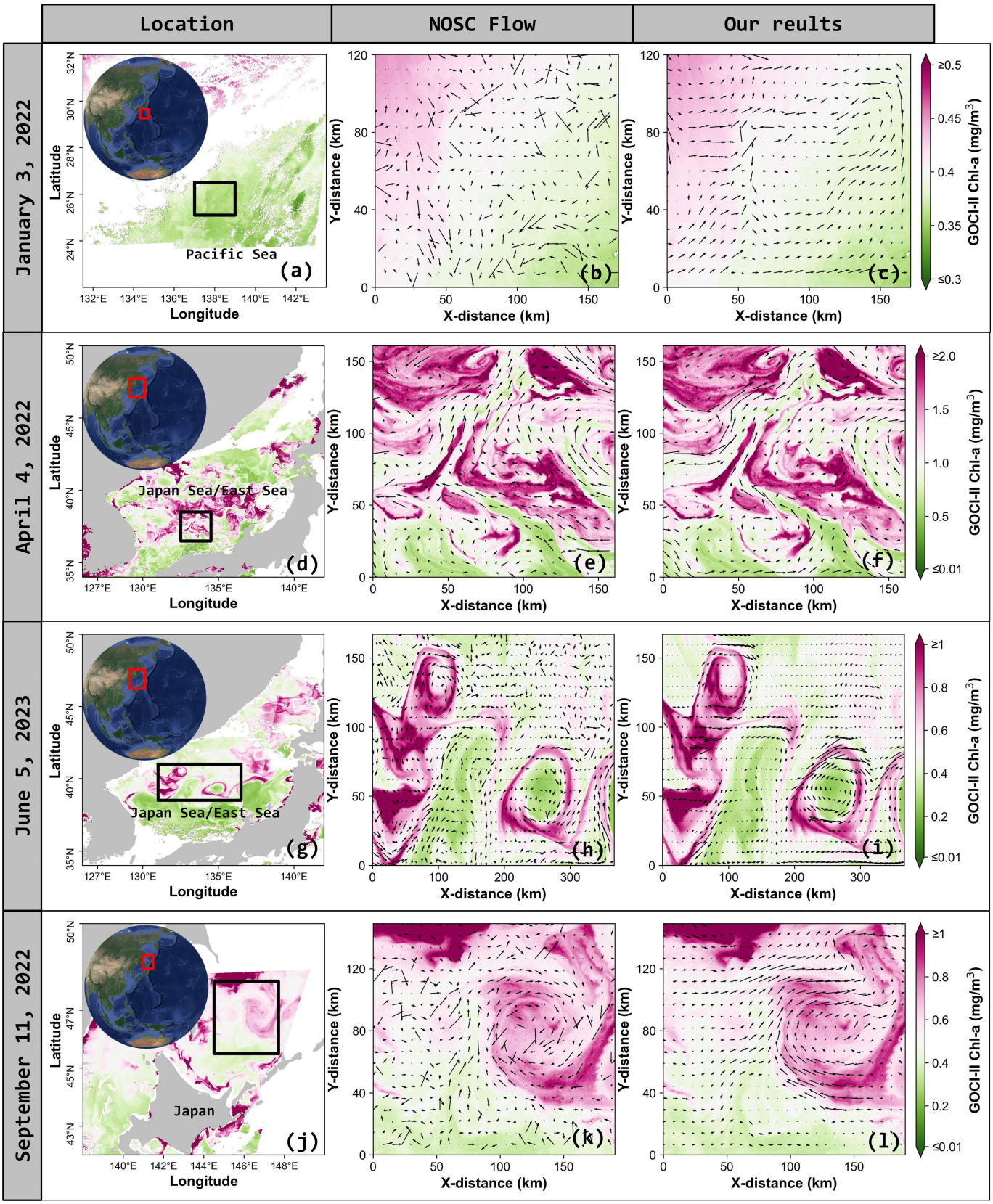


Fig. 4. Daily distribution of mesoscale eddy flow fields from the NOSC product and our deep learning architecture using GOCI-II/GK2B observations.

filamentous Chl-a structure [Fig. 4(k)]. However, the surface flow field derived by the RAFT-MesoEddy architecture was clear and closely matched the filamentous spatial distribution characteristics of phytoplankton [Fig. 4(l)]. Unlike the MCC algorithm, the optical flow algorithm does not require prior adjustment of matching parameters or postprocessing, making it suitable for extracting information from complex flow fields [28], [56].

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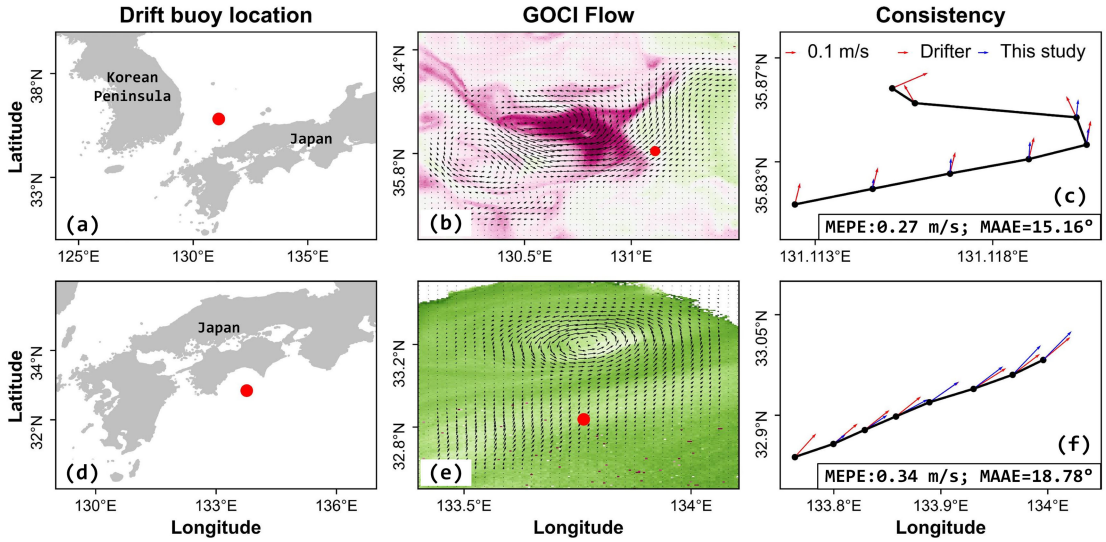


Fig. 5. Comparison of the consistency of the RAFT-MesoEddy architecture retrieved flow field with the two drifting buoys trajectory data on April 5, 2011. (a) and (d) Drifter buoy locations. (b) and (e) Zoomed images of the drifting buoy in the surrounding area, and the daily flow field was derived by the GOCI Chl-a concentration data. (c) and (f) Comparisons of the hourly flow fields between RAFT-MesoEddy architecture retrieved and drifter measured (red), respectively.

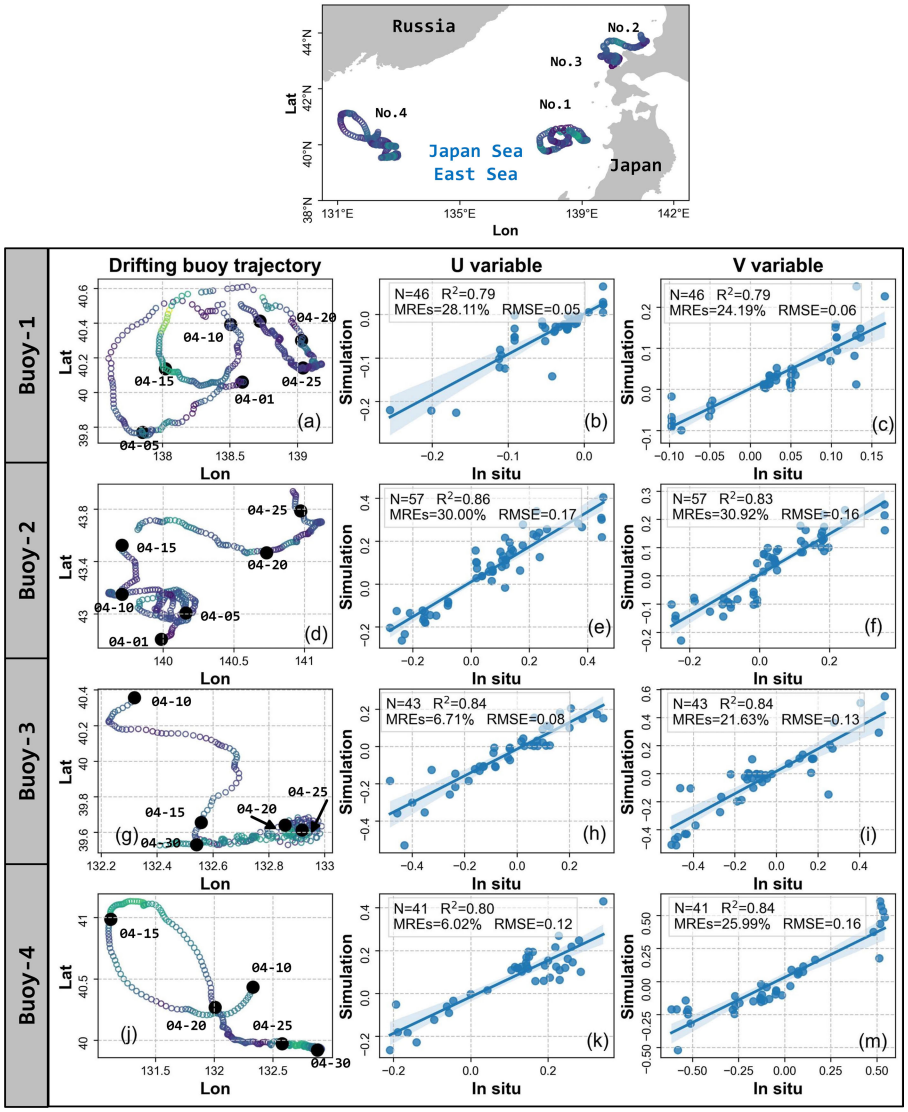


Fig. 6. Comparison of the consistency of the flow field retrieved by the RAFT-MesoEddy architecture with the trajectory data of the four drifting buoys from April 1 to April 30, 2016. (a) Overview of the four drifter locations. (b)–(e), (f)–(i), and (j)–(m) Trajectories, EPE histograms and AEE histograms for the corresponding four drifting buoys, respectively.

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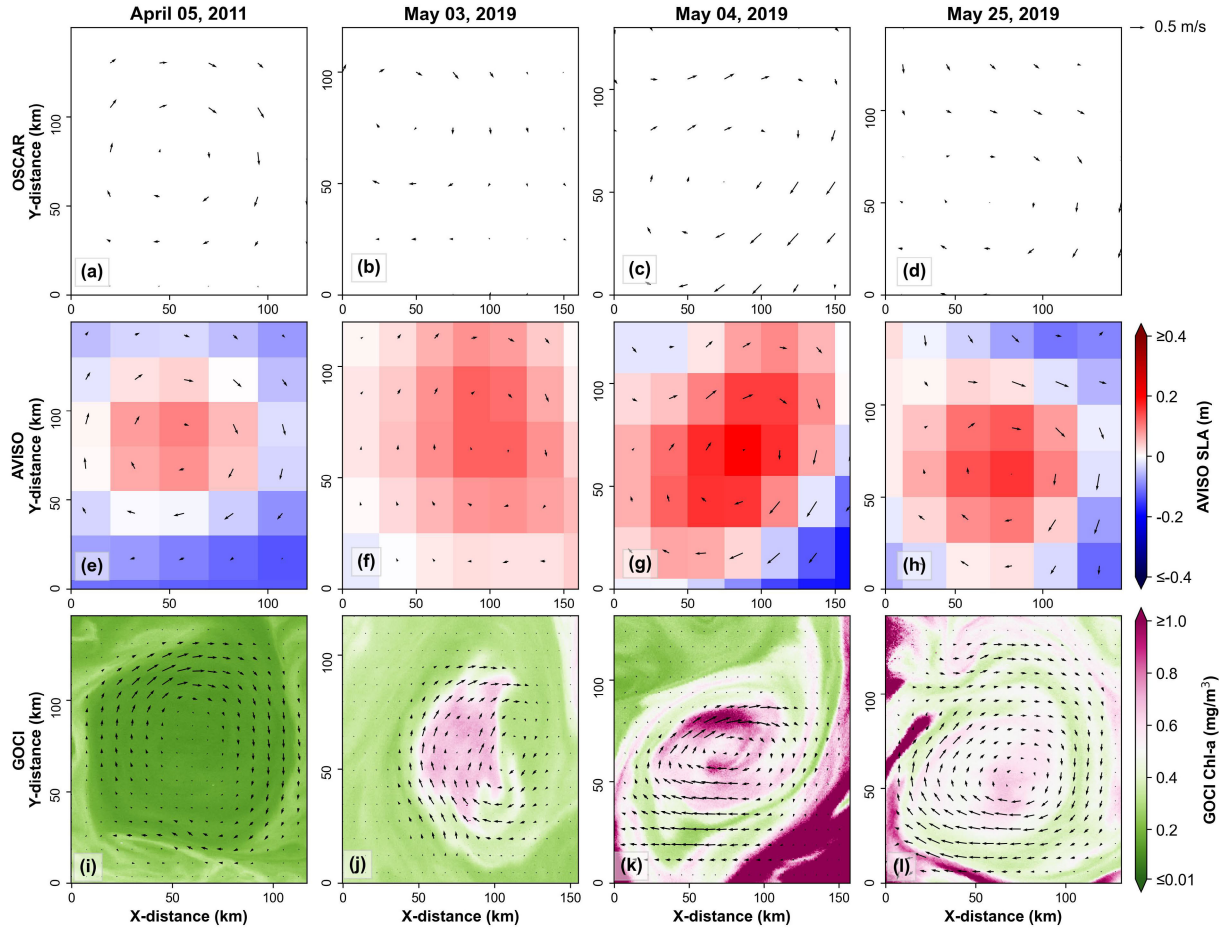


Fig. 7. Daily spatial distribution comparison of different mesoscale processes derived by the OSCAR surface flow field, AVISO geostrophic flow field, and GOCI surface flow field. (a)–(d) Surface flow field from OSCAR data. (e)–(h) Geostrophic flow fields from AVISO data, and the background color are AVISO sea level anomaly (SLA) data. (i)–(l) Surface flow fields from GOCI data, and the background color is the Chl-a concentration.

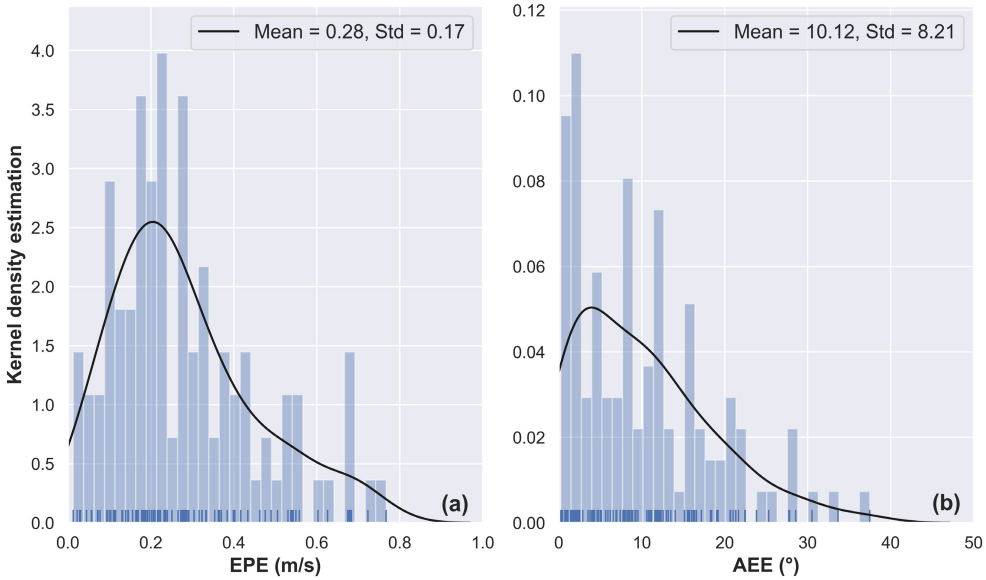


Fig. 8. Histograms of EPE (a) and AEE (b) errors between AVISO geostrophic flow and our retrieval surface flow field. The data are from Fig. 7.

In Section IV-C, we validate our high spatiotemporal resolution flow field derived by the RAFT-MesoEddy architecture using drifting buoy trajectory data, the OSCAR surface flow field, and the AVISO geostrophic flow field, assessing both the quality and spatial distribution.

C. Consistency Between RAFT-MesoEddy Retrieved Flow and Reference Flow From Drift Buoy Trajectory and Satellite Altimeter Data

In the diurnal hourly variations, the GOCI-retrieved flow field and the drifting buoy trajectories data were highly

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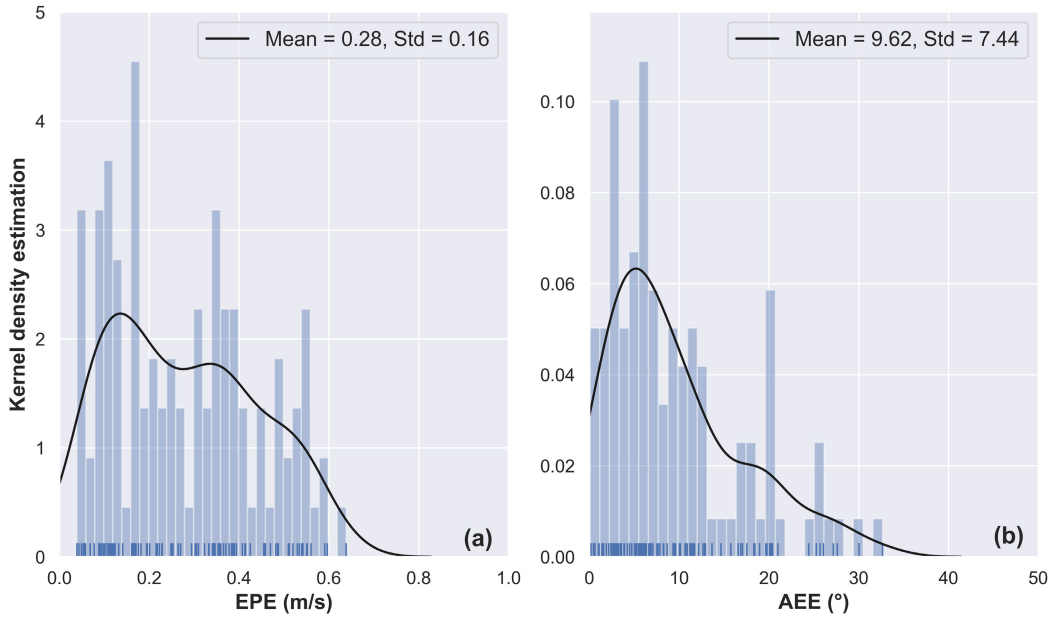


Fig. 9. Histograms of EPE (a) and AEE (b) errors between OSCAR surface flow and our retrieval surface flow field. The data are from Fig. 7.

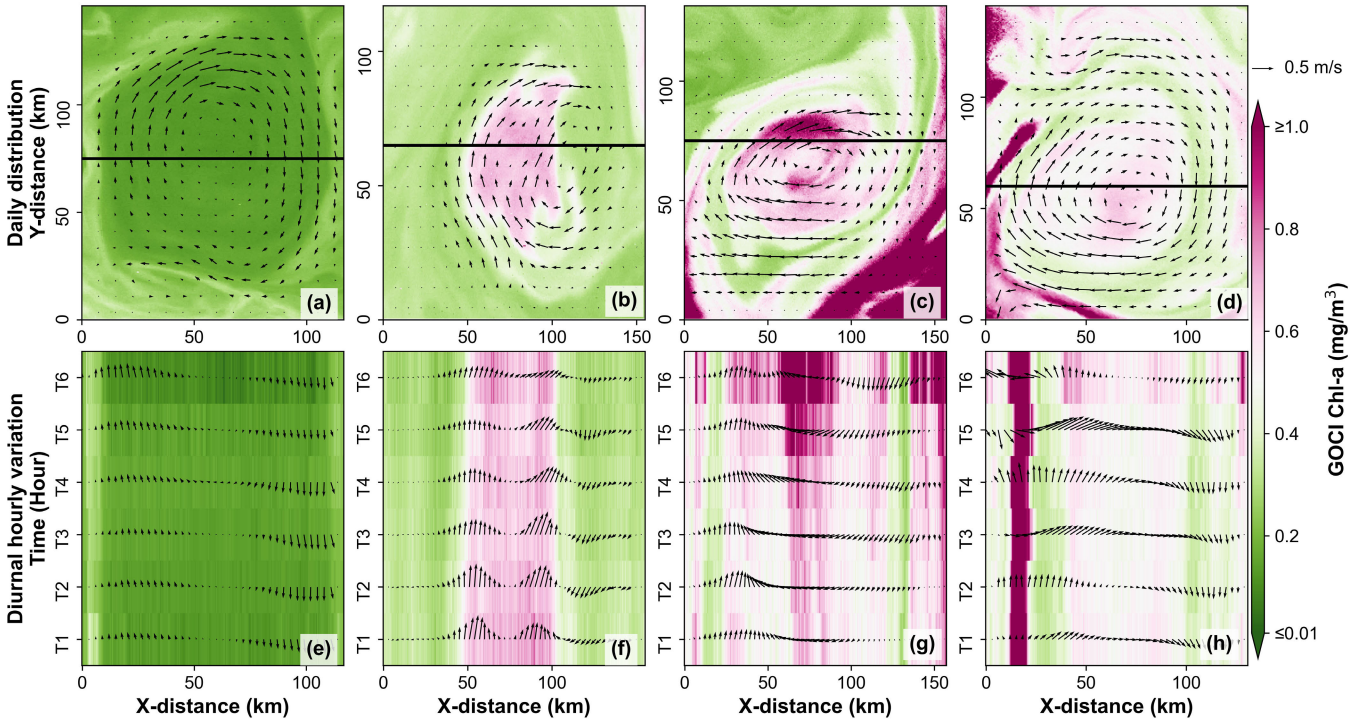


Fig. 10. Diurnal hour flow field and Chl-a concentration distribution pattern of specific sections in the mesoscale eddies. (a)–(d) Daily flow field and Chl-a concentration distribution patterns. (e)–(h) Diurnal hour flow field and Chl-a concentration change pattern in the specific section. $T1$–$T6$ represented specific time intervals ($T1$: 00:15–01:15; $T2$: 01:15–02:15; $T3$: 02:15–03:15; $T4$: 03:15–04:15; $T5$: 04:15–05:15; $T6$: 05:15–06:15).

consistent in diurnal hourly variations, with MEPE of 0.27 [Fig. 5(b)] and 0.34 m/s [Fig. 5(d)], respectively. The flow angle was also consistent with MAAEs of 15.16° [Fig. 5(b)] and 18.78° [Fig. 5(d)], respectively. In Fig. 6, four drifting buoy trajectory data were used to validate the accuracy of the RAFT-MesoEddy architecture from April 1 to April 30, 2016. The GOCI-retrieved flow field and the drifting buoy trajectory data were highly consistent, with MEPE less than 0.20 m/s. The flow angle was also consistent with MAAEs less than 10.0°. Compared to hourly and long-term drifting buoy trajectory data, the RAFT-MesoEddy architecture showed good performance in retrieving mesoscale flow fields from geostationary satellite observations. The EPE and AAE errors of the flow field derived by the MCC algorithm reached up to 0.5 m/s and 31°, respectively, in the Changjiang River estuary [19], [23]. The RAFT-MesoEddy architecture-retrieved

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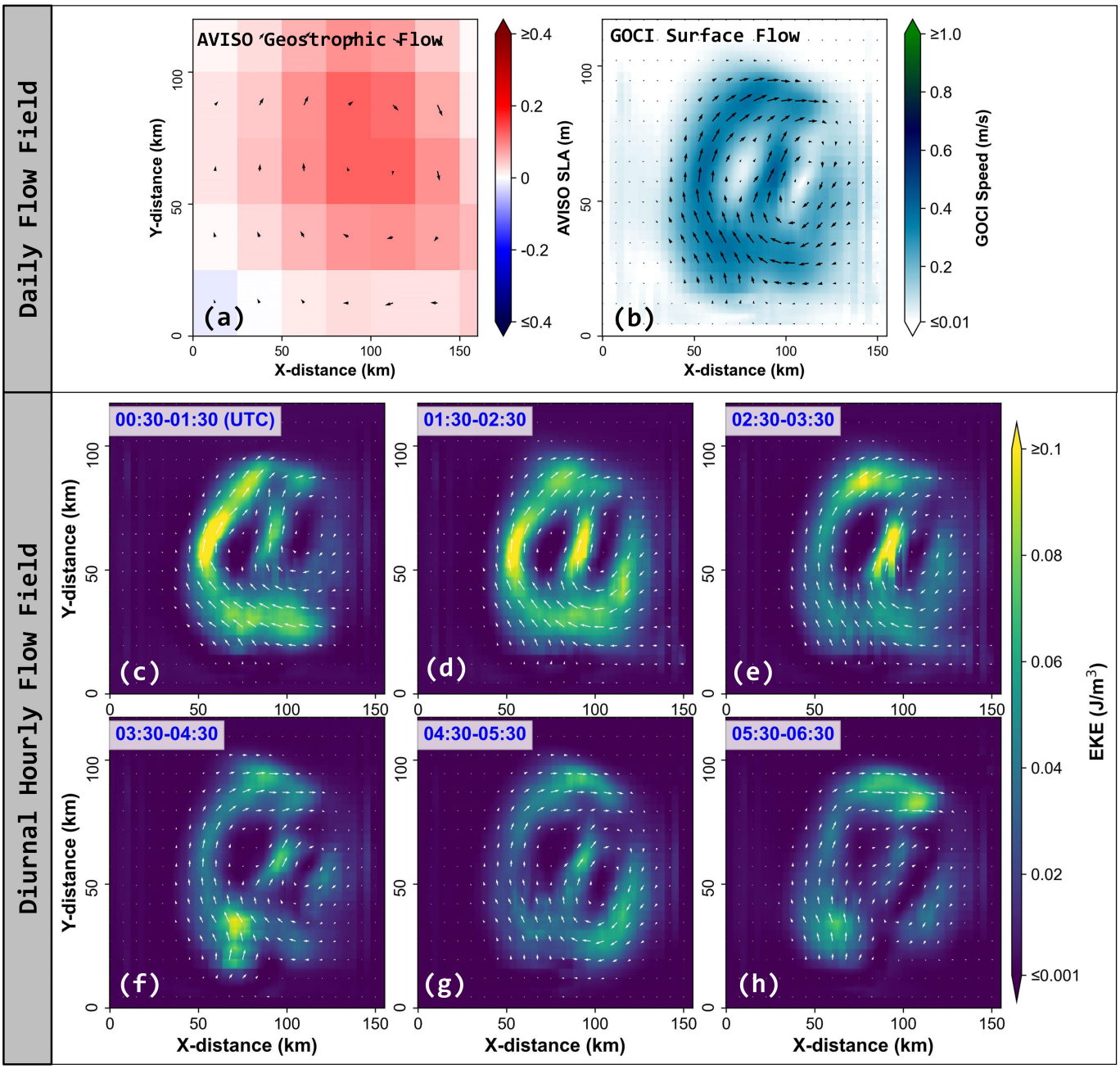


Fig. 11. Daily and diurnal hour flow field and EKE variations in the mesoscale eddy in the May 3, 2019. (a) and (b) AVISO geostrophic flow field and the GOCI surface flow field, respectively. (c)–(h) Diurnal hourly surface flow field and EKE variation.

flow field has better accuracy than the traditional MCC algorithm.

The RAFT-MesoEddy retrieved hourly flow data were averaged into daily flow fields to compare with the OSCAR surface flow field and AVISO geostrophic flow field, as shown in Fig. 7. Yet, the refined GOCI flow field structure facilitated a more precise observation of mesoscale dynamic spatial characteristics [21], [22]. The AVISO data could only resolve relatively coarse spatiotemporal geostrophic surface flow within a limited range of spatiotemporal scales, whereas GOCI provides estimates of the realistic total surface flow field, including both geostrophic and ageostrophic components [54]. The surface flow field derived from GOCI by the RAFT-MesoEddy architecture was clear and closely matched the filamentous spatial distribution pattern of Chl-a concentration.

The performance of the RAFT-MesoEddy architecture was also stable over diurnal hourly variations in Chl-a concentration (Fig. 3). Fig. 8 shows a good agreement of EPE and AEE between the GOCI-retrieved flow field and the AVISO geostrophic flows in four mesoscale eddies, with the MEPE and MAEE being 0.28 m/s and 10.12°, respectively. Both GOCI-retrieved and OSCAR surface flows also showed good agreement, with the MEPE and MAEE being 0.28 m/s and 9.62°, respectively (Fig. 9). Compared to the AVISO and OSCAR data, the uncertainty of the GOCI flow field may

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be closely related to the spatial resolution. When the GOCI flow fields are downsampled for spatial matching, a lot of flow field detail can lead to uncertainty, especially in the eddy boundary regions [48], [56]. These results suggest that the RAFT-MesoEddy architecture can successfully map the diurnal dynamics of the oceanic surface flow field from the GOCI/GOCI-II observations.

D. Demonstrations of the Application of RAFT-MesoEddy Architecture to Monitor Hourly Variations in Eddy Flow Field and Kinetic Energy

High spatiotemporal resolution observation and simulation data provided important data to discuss mesoscale and submesoscale eddy kinetic energy (EKE) transport [51], [58]. In Fig. 10, AVISO presented a geostrophic surface flow field with relatively coarse spatial resolution [Fig. 10(a)], while GOCI offered finer spatiotemporal resolution, illustrating a realistic total surface flow that encompasses both geostrophic and ageostrophic (e.g., tidal and wind-driven flow) components [54], [55]. Compared with the AVISO geostrophic flow field in Fig. 10(a), we observed two regions of minimal velocity in the GOCI image [Fig. 10(b)–(h)], indicating the presence of two small-scale eddies with significant diurnal variations. The center positions of these two eddies generally exhibited parallel distributions during the diurnal hourly changes. When comparing the diurnal hourly EKE at the eddy boundary, the two small-scale anticyclonic eddies showed a tendency to merge into a single mesoscale eddy between 00:30 and 05:30. Specifically, from 00:30 to 03:30, the boundary of the EKE between the two eddy centers was distinct (EKE $\geq$ 0.1 J/m^3). Subsequently, from 03:30 to 05:30, the boundary gradually diminished (EKE $\leq$ 0.08 J/m^3). After 05:30, the EKE at the boundary of the two eddies continued to decrease (EKE $\leq$ 0.04 J/m^3), indicating a tendency for them to merge into a unified mesoscale eddy. The hourly changes in EKE can exhibit amplitudes of up to 25%. Additionally, clear hourly spatial gradients of EKE are observed within the mesoscale eddy, extending from the center position to the boundary regions in both the Northward and Southward areas (Fig. 11). The perspective of diurnal high-frequency observations holds great appeal for physical oceanographers due to the pivotal role played by mesoscale and submesoscale flows in shaping ocean circulation and their interaction within the climate system [59].

In high spatiotemporal observation and simulation, the Surface Water Ocean Topography (SWOT) satellite altimetry mission has the potential to provide high spatial resolution (2 km) observations of global oceanic mesoscale and submesoscale flow fields [59]. The MITgcm provided LLC4320 global ocean simulation product (spatial resolution: 1/48°, and temporal resolution: 1 h) has been employed in several works to understand interactions between the mesoscale and submesoscale turbulence in the upper ocean [51], [58], [60]. The refined flow field findings match those observed in earlier studies [28], [56]; our proposed algorithm is in line with recent studies indicating that diurnal small-scale process can be captured with high spatiotemporal flow field retrieval in the RAFT-MesoEddy architecture. However, the major concerns about the RAFT-MesoEddy architecture findings were missed flow fields affected by cloud coverage, limitations that could be overcome by combining geostationary satellite data and altimeter data in the future.

V. Conclusion

In this study, we demonstrated the robustness of the RAFT-MesoEddy architecture in retrieving sea surface flow fields of oceanic mesoscale processes from geostationary ocean color satellite observations. The flow fields derived from GOCI data aligned well with drifting buoy trajectory data and AVISO geostrophic flow data, illustrating the architecture's strong performance. Notably, the diurnal hourly displacement magnitudes from GOCI data consistently matched the drifting buoy data, with MEPE and MAAE values below 0.34 m/s and 18.78°, respectively. Furthermore, the alignment between the flow fields from GOCI and the data from four drifting buoys was particularly precise, with MEPE and MAAE below 0.20 m/s and 10.0°, respectively. The GOCI-retrieved flow field also showed a good correlation with the AVISO geostrophic flow field, and similarly favorable comparisons were made with OSCAR surface flows.

The mesoscale flow field exhibited consistent continuity across diurnal hourly variations, especially in rotationally distorted flow fields, demonstrating the RAFT-MesoEddy architecture's transferability and robustness with GOCI-II Chl-a products. This technology enhances the study of mesoscale eddy flow fields, offering high temporal resolution and complementing the limitations inherent in AVISO altimeter data. Consequently, this study provides a crucial reference for advancing research into the diurnal dynamics of mesoscale and submesoscale eddies through geostationary ocean color satellite.

However, geostationary ocean color satellite imagery is highly susceptible to cloud cover, leading to significant gaps in the valid data of Chl-a products. These gaps alter the spatial gradient characteristics of tracer particles, introducing substantial uncertainties in the estimation of sea surface flow fields using the RAFT-MesoEddy architecture. Additionally, during dawn and dusk, weak solar illumination conditions caused by large solar zenith angles ($\geq$70) can lead to significant atmospheric correction errors. This not only compromises the accuracy of the sea surface Chl-a products but also results in data loss, causing deviations in the estimated flow fields. Future research should focus on assessing the impact of cloud masking on the refined inversion of flow fields within a deep learning-based optical flow framework. Simultaneously, the selection of sea surface tracer particles must consider the uncertainties in ocean color product accuracy caused by Earth's curvature effect at dawn and dusk.

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REFERENCES

[1] G. Chen, Y. Li, Q. Xie, and D. Wang, "Origins of eddy kinetic energy in the bay of Bengal," *J. Geophys. Res., Oceans*, vol. 123, no. 3, pp. 2097–2115, Mar. 2018, doi: 10.1002/2017jc013455.

[2] R. McCulloch and H. Bindra, "Coupled radiative and conjugate heat transfer in participating media using lattice Boltzmann methods," *Comput. Fluids*, vol. 124, pp. 261–269, Jan. 2016, doi: 10.1016/j.compfluid.2015.05.024.

[3] A. J. Semtner and R. M. Chervin, "Ocean general circulation from a global eddy-resolving model," *J. Geophys. Res., Oceans*, vol. 97, no. C4, pp. 5493–5550, Apr. 1992.

[4] B. A. Storer, M. Buzzicotti, H. Khatri, S. M. Griffies, and H. Aluie, "Global energy spectrum of the general oceanic circulation," *Nature Commun.*, vol. 13, no. 1, p. 5314, Sep. 2022, doi: 10.1038/s41467-022-33031-3.

[5] R. Ferrari and C. Wunsch, "Ocean circulation kinetic energy: Reservoirs, sources, and sinks," *Annu. Rev. Fluid Mech.*, vol. 41, no. 1, pp. 253–282, Jan. 2009, doi: 10.1146/annurev.fluid.40.111406.102139.

[6] D. G. Evans, E. Frajka-Williams, and A. C. N. Garabato, "Dissipation of mesoscale eddies at a western boundary via a direct energy cascade," *Sci. Rep.*, vol. 12, no. 1, p. 887, Jan. 2022, doi: 10.1038/s41598-022-05002-7.

[7] S. Elipot, R. Lumpkin, and G. Prieto, "Modification of inertial oscillations by the mesoscale eddy field," *J. Geophys. Res., Oceans*, vol. 115, no. C9, Sep. 2010, Art. no. C09010, doi: 10.1029/2009jc005679.

[8] J. Choi, Y. G. Park, W. Kim, and Y. H. Kim, "Characterization of submesoscale turbulence in the east/japan sea using geostationary ocean color satellite images," *Geophys. Res. Lett.*, vol. 46, no. 14, pp. 8214–8223, 2019.

[9] Z. Hu et al., "Characterizing surface circulation in the Taiwan strait during NE monsoon from geostationary ocean color imager," *Remote Sens. Environ.*, vol. 221, pp. 687–694, Feb. 2019, doi: 10.1016/j.rse.2018.12.003.

[10] S. Elipot, R. Lumpkin, and R. C. Perez, "A new global surface drifter dataset at hourly resolution," *Amer. Geophys. Union*, vol. 2016, Feb. 2016, Art. no. PO44D.

[11] S. Elipot, A. Sykulski, R. Lumpkin, L. Centurioni, and M. Pazos, "Hourly location, current velocity, and temperature collected from Global Drifter Program drifters world-wide," *Accession*, vol. 248584, p. v1, Jan. 2022.

[12] T. T. M. Hong, Y.-G. Park, and J. M. Choi, "Divergence observation in a mesoscale eddy during Chla Bloom revealed in submesoscale satellite currents," *Remote Sens.*, vol. 15, no. 4, p. 995, Feb. 2023, doi: 10.3390/rs15040995.

[13] C. B. Rocha, S. T. Gille, T. K. Chereskin, and D. Menemenlis, "Seasonality of submesoscale dynamics in the Kuroshio extension," *Geophys. Res. Lett.*, vol. 43, no. 21, p. 11, Nov. 2016.

[14] E. P. Chassignet et al., "The HYCOM (HYbrid Coordinate Ocean Model) data assimilative system," *J. Mar. Syst.*, vol. 65, nos. 1–4, pp. 60–83, Mar. 2007.

[15] A. Holt, J. Boström, G. Karlström, and R. Lindh, "A NEMO potential that includes the dipole–quadrupole and quadrupole–quadrupole polarizability," *J. Comput. Chem.*, vol. 31, no. 8, pp. 1583–1591, Jun. 2010, doi: 10.1002/jcc.21502.

[16] X. Ding et al., "High-frequency and tidal period observations of suspended particulate matter in coastal waters by AHI/Himawari-8," *Opt. Exp.*, vol. 28, no. 19, p. 27387, 2020, doi: 10.1364/oe.401323.

[17] J. Liu, W. J. Emery, X. Wu, M. Li, C. Li, and L. Zhang, "Computing ocean surface currents from GOCI ocean color satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7113–7125, Dec. 2017, doi: 10.1109/TGRS.2017.2741924.

[18] X. He et al., "Using geostationary satellite ocean color data to map the diurnal dynamics of suspended particulate matter in coastal waters," *Remote Sens. Environ.*, vol. 133, pp. 225–239, Jun. 2013, doi: 10.1016/j.rse.2013.01.023.

[19] Z. Hu, L. Li, J. Zhao, and D. Wang, "An objective method with a continuity constraint for improving surface velocity estimates from the geostationary ocean color imager," *Remote Sens.*, vol. 14, no. 1, p. 14, Dec. 2021, doi: 10.3390/rs14010014.

[20] K. Ruddick, G. Neukermans, Q. Vanhellemont, and D. Jolivet, "Challenges and opportunities for geostationary ocean colour remote sensing of regional seas: A review of recent results," *Remote Sens. Environ.*, vol. 146, pp. 63–76, Apr. 2014, doi: 10.1016/j.rse.2013.07.039.

[21] W. Kim, J.-E. Moon, Y.-J. Park, and J. Ishizaka, "Evaluation of chlorophyll retrievals from geostationary ocean color imager (GOCI) for the North-East Asian region," *Remote Sens. Environ.*, vol. 184, pp. 482–495, Oct. 2016, doi: 10.1016/j.rse.2016.07.031.

[22] J. M. Choi, W. Kim, T. T. M. Hong, and Y.-G. Park, "Derivation and evaluation of satellite-based surface current," *Frontiers Mar. Sci.*, vol. 8, Nov. 2021, Art. no. 695780, doi: 10.3389/fmars.2021.695780.

[23] Z. Hu et al., "Assessment of the MCC method to estimate sea surface currents in highly turbid coastal waters from GOCI," *Int. J. Remote Sens.*, vol. 38, no. 2, pp. 572–597, Jan. 2017, doi: 10.1080/01431161.2016.1268737.

[24] Y. Lee and S. Mei, "Diffeomorphic particle image velocimetry," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–10, 2022, doi: 10.1109/Tim.2021.3132999.

[25] L. Alvarez, J. Weickert, and J. Sánchez, "Reliable estimation of dense optical flow fields with large displacements," *Int. J. Comput. Vis.*, vol. 39, no. 1, pp. 41–56, 2000, doi: 10.1023/a:1008170101536.

[26] J. Walker, A. Gupta, and M. Hebert, "Dense optical flow prediction from a static image," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2443–2451.

[27] J. Wulff and M. J. Black, "Efficient sparse-to-dense optical flow estimation using a learned basis and layers," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 120–130.

[28] P. Dérian and R. Almar, "Wavelet-based optical flow estimation of instant surface currents from shore-based and UAV videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5790–5797, Oct. 2017, doi: 10.1109/TGRS.2017.2714202.

[29] E. Ilg, N. Mayer, T. Saikia, M. Keuper, A. Dosovitskiy, and T. Brox, "FlowNet 2.0: Evolution of optical flow estimation with deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2462–2470.

[30] D. Sun, X. Yang, M.-Y. Liu, and J. Kautz, "PWC-Net: CNNs for optical flow using pyramid, warping, and cost volume," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8934–8943.

[31] T.-W. Hui, X. Tang, and C. C. Loy, "LiteFlowNet: A lightweight convolutional neural network for optical flow estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8981–8989.

[32] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *Proc. Eur. Conf. Comput. Vis.*, Glasgow, U.K. Cham, Switzerland: Springer, Aug. 2020, pp. 402–419.

[33] S. Cai, S. Zhou, C. Xu, and Q. Gao, "Dense motion estimation of particle images via a convolutional neural network," *Exp. Fluids*, vol. 60, no. 4, pp. 1–16, Apr. 2019.

[34] M. Zhang and M. D. Piggott, "Unsupervised learning of particle image velocimetry," in *Proc. Int. Conf. High Perform. Comput.* Frankfurt, Germany. Cham, Switzerland: Springer, 2020, pp. 102–115.

[35] C. Lagemann, K. Lagemann, S. Mukherjee, and W. Schröder, "Deep recurrent optical flow learning for particle image velocimetry data," *Nature Mach. Intell.*, vol. 3, no. 7, pp. 641–651, Jul. 2021, doi: 10.1038/s42256-021-00369-0.

[36] D. Sun, C. Herrmann, F. Reda, M. Rubinstein, D. J. Fleet, and W. T. Freeman, "Disentangling architecture and training for optical flow," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 165–182.

[37] A. Jahedi, L. Mehl, M. Rivinius, and A. Bruhn, "Multi-scale raft: Combining hierarchical concepts for learning-based optical flow estimation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 1236–1240. [Online]. Available: https://ieeexplore.ieee.org/document/9898048/

[38] J. E. O'Reilly et al., "Ocean color chlorophyll algorithms for SeaWiFS," *J. Geophys. Res., Oceans*, vol. 103, no. C11, pp. 24937–24953, Oct. 1998, doi: 10.1029/98jc02160.

[39] J. Nasa, *Ocean Surface Current Analyses Real-Time (OSCAR) Surface Currents-Interim 0.25 Degree (Version 2.0)*, OSCAR_L4_OC_INTERIM_V2. 0, Jet Propulsion Laboratory Physical Oceanography, Pasadena, CA, USA, 2020.

[40] J.-E. Park, K.-A. Park, D. S. Ullman, P. C. Cornillon, and Y.-J. Park, "Observation of diurnal variations in mesoscale eddy sea-surface currents using GOCI data," *Remote Sens. Lett.*, vol. 7, no. 12, pp. 1131–1140, Dec. 2016, doi: 10.1080/2150704x.2016.1219423.

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[41] J.-E. Park, K.-A. Park, C.-K. Kang, and Y.-J. Park, "Short-term response of Chlorophyll–A concentration to change in sea surface wind field over mesoscale eddy," *Estuaries Coasts*, vol. 43, no. 3, pp. 646–660, May 2020.

[42] A. Doshi and A. G. Bors, "Smoothing of optical flow using robustified diffusion kernels," *Image Vis. Comput.*, vol. 28, no. 12, pp. 1575–1589, Dec. 2010, doi: 10.1016/j.imavis.2010.04.001.

[43] T. Käufer, J. König, and C. Cierpka, "Stereoscopic PIV measurements using low-cost action cameras," *Experim. Fluids*, vol. 62, no. 3, pp. 1–16, Mar. 2021.

[44] G. Farnebäck, "Two-frame motion estimation based on polynomial expansion," in *Proc. Image Anal., 13th Scand. Conf.* Switzerland: Springer, 2003, pp. 363–370, doi: 10.1007/3-540-45103-X_50.

[45] X. Ding et al., "Requirement on the diurnal observation frequency for satellite remote sensing of photosynthetically available radiation," *Opt. Exp.*, vol. 30, no. 6, p. 9021, Mar. 2022, doi: 10.1364/oe.451207.

[46] Z. Song et al., "Atmospheric correction of absorbing aerosols for satellite ocean color remote sensing over coastal waters," *Remote Sens. Environ.*, vol. 290, May 2023, Art. no. 113552, doi: 10.1016/j.rse.2023.113552.

[47] Y. Xu and J. Chen, "Uncertainty estimation in geostationary ocean color imager (GOCI) bio-optical products for detecting diurnal variability in coastal water," *Int. J. Remote Sens.*, vol. 44, no. 5, pp. 1484–1509, Mar. 2023, doi: 10.1080/01431161.2023.2185113.

[48] H. Yang, J.-K. Choi, Y.-J. Park, H.-J. Han, and J.-H. Ryu, "Application of the geostationary ocean color imager (GOCI) to estimates of ocean surface currents," *J. Geophys. Res., Oceans*, vol. 119, no. 6, pp. 3988–4000, Jun. 2014, doi: 10.1002/2014jc009981.

[49] P.-C. Hsu, C.-Y. Ho, H.-J. Lee, C.-Y. Lu, and C.-R. Ho, "Temporal variation and spatial structure of the Kuroshio-induced submesoscale island vortices observed from GCOM-C and Himawari-8 data," *Remote Sens.*, vol. 12, no. 5, p. 883, Mar. 2020, doi: 10.3390/rs12050883.

[50] E. Carli, R. Morrow, O. Vergara, R. Chevrier, and L. Renault, "Ocean 2D eddy energy fluxes from small mesoscale processes with SWOT," *Ocean Sci.*, vol. 19, no. 5, pp. 1413–1435, Oct. 2023, doi: 10.5194/os-19-1413-2023.

[51] J. Dong, B. Fox-Kemper, H. Zhang, and C. Dong, "The seasonality of submesoscale energy production, content, and cascade," *Geophys. Res. Lett.*, vol. 47, no. 6, Mar. 2020, Art. no. e2020GL087388, doi: 10.1029/2020gl087388.

[52] J.-H. Ahn, Y.-J. Park, J.-H. Ryu, B. Lee, and I. S. Oh, "Development of atmospheric correction algorithm for geostationary ocean color imager (GOCI)," *Ocean Sci. J.*, vol. 47, no. 3, pp. 247–259, Sep. 2012, doi: 10.1007/s12601-012-0026-2.

[53] J. Choi, Y. J. Park, J. H. Ahn, H. Lim, J. Eom, and J. Ryu, "GOCI, the world's first geostationary ocean color observation satellite, for the monitoring of temporal variability in coastal water turbidity," *J. Geophys. Res., Oceans*, vol. 117, no. C9, Sep. 2012, Art. no. C09004, doi: 10.1029/2012jc008046.

[54] Z. Hu, H. Zhang, and D. Wang, "A novel approach for estimating sea surface currents from numerical models and satellite images: Validation and application," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3370996.

[55] F. Ye, Z. Hao, and D. Pan, "An optimization method based on decorrelation scales analysis for improving surface currents retrieval from sea surface temperature," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/TGRS.2024.3360512.

[56] H. Sun, Q. Song, R. Shao, and T. Schlicke, "Estimation of sea surface currents based on ocean colour remote-sensing image analysis," *Int. J. Remote Sens.*, vol. 37, no. 21, pp. 5105–5121, Nov. 2016, doi: 10.1080/01431161.2016.1226526.

[57] J. G. Yoo, S. Y. Kim, and H. S. Kim, "Spectral descriptions of submesoscale surface circulation in a coastal region," *J. Geophys. Res., Oceans*, vol. 123, no. 6, pp. 4224–4249, Jun. 2018, doi: 10.1029/2016jc012517.

[58] H. Khatri, S. M. Griffies, T. Uchida, H. Wang, and D. Menemenlis, "Role of mixed-layer instabilities in the seasonal evolution of eddy kinetic energy spectra in a global submesoscale permitting simulation," *Geophys. Res. Lett.*, vol. 48, no. 18, Sep. 2021, Art. no. e2021GL094777, doi: 10.1029/2021gl094777.

[59] S. Metref et al., "Reduction of spatially structured errors in wide-swath altimetric satellite data using data assimilation," *Remote Sens.*, vol. 11, no. 11, p. 1336, Jun. 2019, doi: 10.3390/rs11111336.

[60] C. B. Rocha, T. K. Chereskin, S. T. Gille, and D. Menemenlis, "Mesoscale to submesoscale wavenumber spectra in Drake Passage," *J. Phys. Oceanogr.*, vol. 46, no. 2, pp. 601–620, Feb. 2016, doi: 10.1175/JPO-D-15-0087.1.



Xiaosong Ding received the M.S. degree in ocean science from Shanghai Ocean University, Shanghai, China, in 2019, and the Ph.D. degree in physical oceanography from Hohai University, Nanjing, China, in 2023.

He is currently a Post-Doctoral Research Associate with Donghai Laboratory, Zhoushan, China, and Zhejiang University, Zhoushan. His research interests include ocean color remote sensing, geostationary satellite observation, artificial intelligence oceanography, satellite data analysis, and data assimilation.



Xianqiang He received the B.S. degree in marine and ocean engineering from Huazhong University of Science and Technology, Wuhan, China, in 1999, the M.S. degree in physical oceanography from the Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, China, in 2002, and the Ph.D. degree in physical electronics from Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, China, in 2007.

He is currently with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources. He is also with the School of Oceanography, Shanghai Jiao Tong University, Shanghai. His research interests include radiative transfer in the coupled ocean and atmosphere systems, atmospheric correction of satellite ocean color remote sensing, and oceanography research using ocean color remote sensing data.



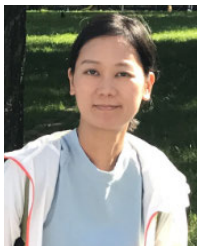
Yan Bai received the B.S. and M.S. degrees from Sun Yat-sen University, Guangzhou, China, in 2000 and 2003, respectively, and the Ph.D. degree in electromagnetic fields and microwave techniques from Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai, China, in 2007.

She is currently with the State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, China, and Zhejiang University, Zhoushan, China. Her research interests include satellite remote sensing of marine carbon cycle.



Jiajia Li received the M.S. degree in ocean science from Shanghai Ocean University, Shanghai, China, in 2021. She is currently pursuing the Ph.D. degree in geography with Yunnan University, Kunming, China.

Her research interests include artificial intelligence geography, satellite data analysis, and data assimilation.



Fang Gong received the B.S. degree in computer and application from Nanjing University of Information Science and Technology, Nanjing, China, in 2001.

She is currently a Senior Engineer with the State Key Laboratory of Satellite Ocean Environment Dynamics (SOED), Second Institute of Oceanography (SIO), Ministry of Natural Resources (MNR), Hangzhou, China. Her research interests include satellite data processing and analysis for ocean environment monitoring in the open ocean and near shore waters.

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